

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Sixth Semester B.E. (Mechanical) (Credit System) Degree Examinations

May - June 2023

20MEE41 – RENEWABLE SOURCES OF ENERGY

Duration: 3 Hours

Max. Marks: 100

Note: 1) Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

2) Use of solar data sheet may be permitted.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the following with the help of sketches				
	i) Latitude of location				
	ii) Declination angle				
	iii) Hour Angle				
	iv) Altitude Angle				
	v) Solar Azimuth angle	10	L*1	1	1
	b) Calculate the hour angle at sunrise and sunset, and the day length on June 21 and December 21 for a surface inclined at an angle of 10° and facing due south. The surface is located in Bombay (19° 07'N, 72° 51' E)	10	L3	1	2
2.	a) Explain the principle of conversion of solar energy into heat. Compare the relative merits and demerits of concentrating collector with flat plate collector.	10	L2	1	1
	b) Calculate the heat storage density for Glauber's salt when its temperature is raised from 27°C to 37°C. Glauber salt has the following properties Melting temperature.....32°C Specific heat (solid).....1926 J/kg°C Specific heat (liquid).....2846 J/kg°C Latent heat2.4x10 ⁵ J/kg Density.....1600kg/m ³ Also calculate the volume required to store 25kW-hr energy and determine the percentage energy store by latent heat.	10	L3	2	2
3.	a) With the help of suitable sketch explain the following				
	i) Solar space heating				
	ii) Solar pond	10	L2	2	1
	b) Explain the principle of chemical energy storage. Explain with a neat diagram of a power plant having a chemical energy storage system utilizing the reaction $\text{CH}_4 + \text{H}_2\text{O} = \text{CO} + 3\text{H}_2$	10	L2	2	1
Unit – II					
4.	a) Draw a neat sketch of Darrieus rotor type and Savonius rotor type VAWT machine. Compare their relative merits and demerits.	10	L2	3	1

- b) Calculate the volume and size of a biogas digester suitable for the output of five cows from the following data

Dry matter produced per cow is 2kg per day

Biogas yield 0.24 m³ per kg

Burner efficiency is 60%

Methane proportion is 0.8

Heat of combustion of methane is 28MJ/m³

Retention time is 20 days

Density of the dry material in the fluid is 50 kg/m³

Total volume of the digester 10% more than the minimum working volume

Cylindrical digester with height to depth ratio is unity.

Also calculate the power available from the digester.

10 L3 3 2

5. a) Write a short note on the following

i) Utilization of biogas in IC engines.

ii) Down draft gasifier

10 L1 3 1

- b) Discuss the different tidal power schemes and configurations with neat sketches.

10 L2 4 1

6. a) What are the advantages and disadvantages of wave energy conversion type energy source?

10 L1 4 1

- b) Explain the basic principle of ocean thermal energy conversion? Discuss the different methods of harnessing ocean thermal energy.

10 L2 4 1

Unit – III

7. a) Explain with neat schematic and T-s diagram vapor dominated geothermal power plant.

10 L2 5 1

- b) What are the possible sources of geothermal pollution? How these are avoided?

10 L1 5 1

8. a) What is a fuel cell? How is it different from a battery?

4 L1 5 1

- b) State the advantages, disadvantages and applications of fuel cells.

8 L2 5 1

- c) A hydrogen-oxygen cell operates at 25°C.

Given: $\Delta H^\circ_{298K} = -285838$ KJ/kg mole., $\Delta G^\circ_{298K} = -237191$ kJ/kg mole

Calculate:

(i) voltage output of the cell

(ii) Efficiency of the fuel cell.

(iii) Electrical work output per mole of H₂ consumed and per mole of H₂O produced.

(iv) Heat transfer to the surroundings.

8 L3 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
